A Functional Preservation Score for Auditing Connectome Operations: Compression, Alignment, Transport, and Local-Rule Writing

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Abstract

We introduce the Functional Preservation Score (FPS), a null-normalised, behaviourally grounded measure of how much of a neural network's input–output function survives a structural operation, and use it to audit the operations underlying speculative "mind-migration" pipelines on public data that runs on a single laptop. Across five pre-registered studies on the *Caenorhabditis elegans* connectome, whole-brain calcium activity, and hippocampal replay, FPS yields two positive characterisations and a three-part negative. Low-bit quantisation of a real connectome exhibits a structure-dependent sparsity floor: uniform int4/ternary quantisation is forced to roughly 73%/98% effective sparsity because heavy-tailed integer synaptic weights round to zero, while mixed-precision protection of hub neurons lowers these floors to about 27%/32%. Decoded hippocampal replay sequences compress heavily yet remain identifiable at a cross-validated top-1 accuracy of 0.69 (95% CI $[0.63, 0.74]$; chance 0.125; permutation $p < 10^{-3}$) from latents of order 10–50 bits per event, and a bounded incremental store shows a graceful-versus-cliff capacity trade-off. The migration pipeline fails in simulation at each stage for largely independent reasons: the putative "identity payload" is not a functionally separable module ($p \geq 0.16$ at every tested size); cross-individual functional transport via Gromov–Wasserstein collapses once the target's wiring is respected (fidelity $1.00 \to 0.40$, $[0.31, 0.49]$, $p < 10^{-3}$) or alignment is imperfect (0.06, $[0.00, 0.13]$); and biologically plausible local plasticity cannot reach the function the target topology provably admits (gap 0.45, $[0.27, 0.64]$, $p < 10^{-3}$, $d = 2.0$). We release FPS, the data loaders, and all operations as a toolkit. All results are at nematode / small-network scale and concern coherence in simulation, not the feasibility of reading or writing a living brain.

1 Introduction

Structural operations on neural networks — pruning, quantisation, alignment across instances, and activity-driven weight installation — appear across model compression, connectomics, and speculative proposals to "migrate" a mind from one substrate to another. Such proposals combine connectomics, graph neural networks, memory prosthetics, neural interfaces, and optogenetics into a single partition–align–write pipeline, typically relying on the most favourable reading of each component. Evaluating these operations requires a common question: how much of a network's *function* does an operation preserve? We make that question measurable with the Functional Preservation Score (FPS) and apply it, on public data that runs on a laptop, both to operations that are genuinely useful (compression) and to the stages of the migration pipeline that lack empirical support.

Our contributions are: (1) FPS, a substrate-agnostic functional-preservation measure with an explicit trivial-baseline null (§3); (2) a characterisation of connectome compression, including a quantisation sparsity floor (§4); (3) a characterisation of replay-sequence compressibility and a capacity–interference relationship (§5); (4) a three-stage in-silico test of the partition–align–write pipeline that localises the failure mode at each stage (§6); and (5) a reproducible toolkit (§8). The unifying technical object is FPS used as a common currency across otherwise disparate operations.

2 Related work

Generic graph-network pruning and quantisation (Khedri et al., 2025) and scalable Gromov–Wasserstein alignment (Xu et al., 2019b;a; Peyré et al., 2016) are established. Brain alignment via optimal transport has been demonstrated at the scale of functional parcels (Thual et al., 2022). Connectome-driven models can reproduce behaviour once dynamical parameters are fitted (Dorkenwald et al., 2024; FlyWire-GNN authors, 2025). The lottery-ticket literature (Frankle & Carbin, 2019) studies sparse trainable subnetworks. Our contribution is not a new compression or transport algorithm but a functional evaluation of these operations on connectome substrates, and an explicit test of migration stages that are frequently asserted but, to our knowledge, never measured. We summarise the biological claim provenance (synaptic turnover (von der Ohe et al., 2006); memory prosthetics (Hampson et al., 2018); surface neural interfaces (Jung et al., 2025)) in Appendix A.

3 The Functional Preservation Score

Let a system map a fixed battery of stimuli to outputs. For a reference system with outputs Y_{ref} and a transformed system with outputs Y, let $\text{raw}(Y) = 1 - \|Y - Y_{\text{ref}}\|^2 / \|Y_{\text{ref}} - \overline{Y_{\text{ref}}}\|^2$. FPS normalises this by the best trivial baseline Y_{null}, the larger of a shuffled stimulus–response mapping and a predict-the-mean baseline:

$$\text{FPS} = \text{clip}\left(\frac{\text{raw}(Y) - \text{raw}(Y_{\text{null}})}{1 - \text{raw}(Y_{\text{null}})},\, 0,\, 1 \right). \tag{1}$$

An FPS of 1 reproduces the reference function; 0 is no better than the best trivial control. Normalising by the null prevents an input-independent but plausible-looking output from scoring well. For identity-style tasks we use identification accuracy (is reconstruction i nearest its own original among all events?), which is bounded with exact chance $1/n$.

All headline numbers are reported as a point estimate with a percentile bootstrap 95% CI (10^4 resamples) over the relevant randomisation; comparisons use one-sided permutation tests (10^4 permutations, add-one-corrected) and Cohen's d where applicable. To avoid selection bias we report a pre-registered primary policy rather than a maximum over variants, and the identification frontier uses 30-split repeated cross-validation with models fit on the training split only.

4 Study 1: Connectome compression

We load the real *C. elegans* somatic connectome (White et al., 1986; Cook et al., 2019) (306 neurons; 81 sensory, 95 interneuron, 130 motor) as a leaky-rate recurrent model and score the motor readout under compression. At matched effective sparsity (int4), the pre-registered engram-aware policy preserves function better than uniform pruning: FPS 0.30 ($[0.12, 0.48]$) versus 0.03 ($[0.00, 0.09]$) at target sparsity 0.5 ($p = 0.015$, $d = 1.3$). The robust result is the floor (Figure 1): uniform quantisation cannot reach low sparsity because the heavy-tailed integer weights round to zero, giving an effective-sparsity floor of 0.73 (int4) and 0.98 (ternary), against 0.27 and 0.32 under engram-aware mixed precision. These floors are deterministic properties of the weight distribution. The structured advantage itself is regime-dependent and does not reproduce under an independent sigmoid-conductance model, so we retain only the floor claim.

5 Study 2: Replay-sequence compression

On hippocampal replay data (DANDI 000044 (Grosmark & Buzsáki, 2016; Diba & Buzsáki, 2007); 48 pyramidal units), place-ordered sequences carry decodable structure (Bayesian decode 0.34 versus a time-shuffled null 0.20). Under 30-split cross-validation, an uncompressed autoencoder identifies held-out events at top-1 0.69 ($[0.63, 0.74]$) against chance 0.125 ($p = 10^{-4}$), degrading gradually with the bit budget (Figure 2, left): a 48×25 sequence ($\sim$ 38 kbit) remains well above chance down to 8–16 bits per event ($\sim 10^3\times$ compression). A fixed-size incremental store (Figure 2, right) shows a genuine capacity limit, graceful for a superposition store and cliff-like for a slot store.

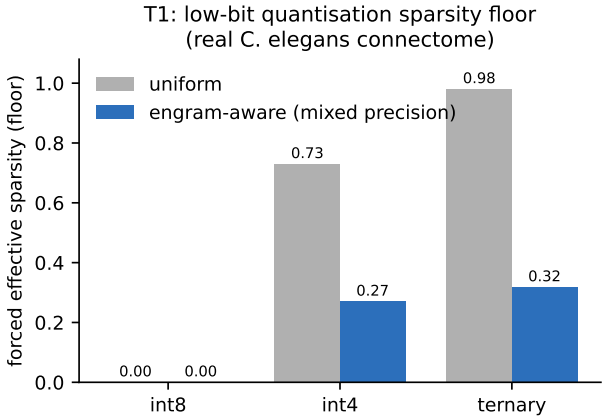


Figure 1: Effective-sparsity floor by bit depth on the real connectome: uniform quantisation (grey) versus engram-aware mixed precision (blue).

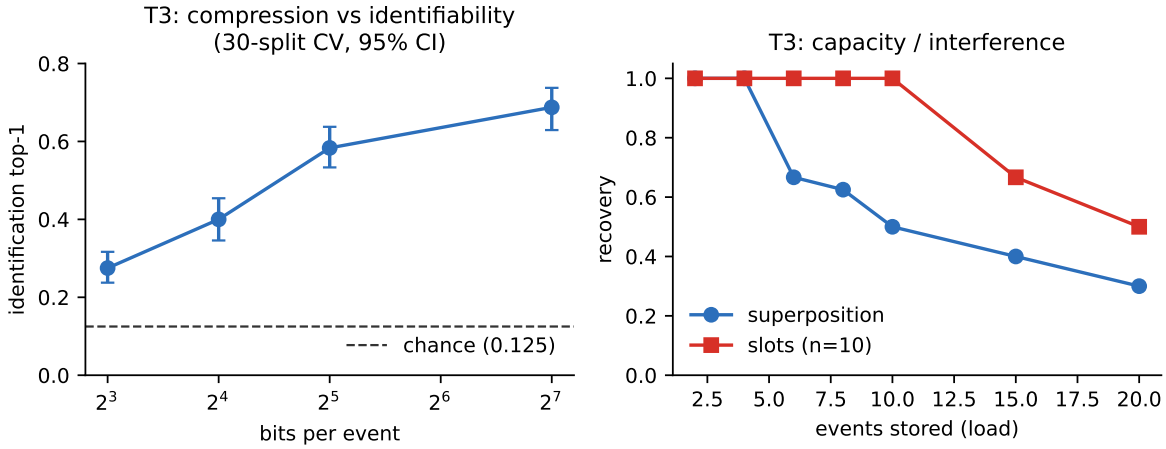


Figure 2: Left: identification versus latent bit budget (30-split CV, 95% CI; dashed line is chance). Right: recovery from a fixed-size store versus load.

6 Study 3: The migration pipeline in simulation

Each stage tests a biological stage for which no direct evidence exists; we ask only whether it is coherent in simulation. **Partition.** Holding the sensory/motor interface fixed, ablating the asserted interneuron payload is no more disruptive than ablating a random interneuron subset of the same size ($p = 0.17, 0.16, 0.27, 0.37$ across payload fractions 0.1–0.4; 500-sample null): there is no significantly separable identity module. **Transport.** Entropic Gromov–Wasserstein recovers 0.76 ($[0.74, 0.78]$) of a ground-truth correspondence at 5% divergence, but transport fidelity collapses from 1.00 (oracle, unconstrained) to 0.40 ($[0.31, 0.49]$, $p = 3 \times 10^{-4}$) once the target's wiring is respected, and to 0.06 ($[0.00, 0.13]$, $p = 2 \times 10^{-4}$) under the estimated correspondence (Figure 3, left). **Writing.** A global optimiser reaches ceiling 1.00 on the target topology, while strictly local reward-modulated plasticity reaches 0.55 ($[0.36, 0.73]$), a plausibility gap of 0.45 ($[0.27, 0.64]$, $p = 10^{-4}$, $d = 2.0$), with ripple timing conferring no advantage (Figure 3, right). The stages fail for independent reasons; the target substrate's wiring is the limit common to transport and writing.

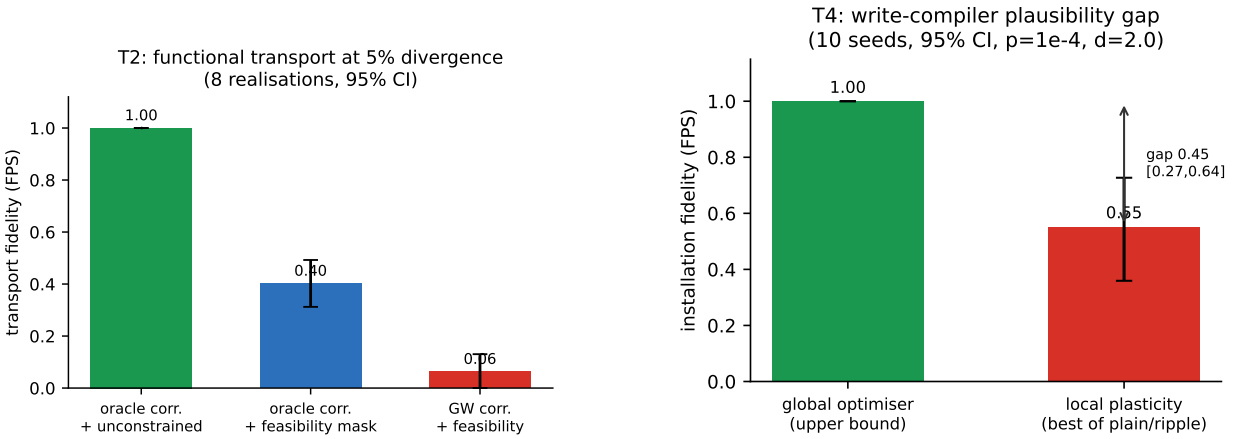


Figure 3: Left: transport fidelity at 5% divergence under three conditions (95% CI). Right: the write-compiler plausibility gap between a global optimiser and local plasticity.

7 Discussion and limitations

FPS cleanly separates the pipeline: compression of neural data is measurable and bounded, whereas migration of function across substrates is unsupported by the partition–align–write pipeline, with the target substrate's wiring as the recurring obstacle. Limitations are material: small scale (a 306-neuron connectome, $\leq$135-neuron activity, 40-neuron networks); simple rate dynamics; small test sets in places; a divergence model rather than a second organism in Study 2; and FPS as a functional measure rather than a claim about subjective experience. The biological prerequisites for the full pipeline are absent and not addressed.

8 Reproducibility

All results are deterministic and run on a laptop. Each claim maps to a committed result artifact and the script that produced it; figures are generated from those artifacts. Statistically hardened artifacts (bootstrap CIs, permutation p-values) accompany each headline claim. Code, data loaders, and the full operation set are released as open source; an anonymised repository is provided for review, with the public repository to be linked in the camera-ready.

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A Biological claim provenance

Torpor sheds ∼50–65% of synapses with memory preserved (von der Ohe et al., 2006), supporting structured rather than uniform sparsification. Memory prosthetics (Hampson et al., 2018) restore recall within a subject, not across subjects. High-density wireless neural interfaces (Jung et al., 2025) record at the subdural surface, not synaptic resolution. Whole-brain calcium imaging (Kato et al., 2015) and the signal-propagation atlas (Randi et al., 2023) provide the functional substrates used here. None of these supports the non-destructive synaptic-resolution reading, identity-scale writing, or cross-subject transfer the full pipeline would require.